

NFL Salary Allocation vs. On-Field Performance Analysis

Team Members

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Data Sources

Primary Dataset: NFL Salary by Position Group (2013-2022) – team-level salary spending by position group per season (.csv)

Secondary Dataset: NFL scores and betting data – game-level results with weekly matchups and final scores (.csv)

Project Overview

This project investigates how NFL team salary allocation across position groups relates to on-field performance at the game level.

We will construct a team-game modeling dataset, where each row represents a single team in a single game. Season-level salary allocation will be merged onto each team-game observation to analyze how financial strategy influences individual game outcomes. Specifically, we will:

- Clean and merge salary-by-position data with game results
- Convert each NFL game into two team-level observations (home and away)
- Engineer game-level performance metrics (win/loss, points scored, point differential, home/away indicators)
- Train scikit-learn models to predict game outcomes based on salary distribution
- Deliver an interactive dashboard to explore insights (by team, season, and position group)

Key Objectives

1. Data wrangling + integration

- Filter both datasets to seasons (2013–2022)
- Transform each game into two team-level rows (home team and away team)
- Create game-level outcome variables
 - Win (binary)
 - Points scored
 - Points allowed
 - Point differential
- Merge season-level salary allocation data onto each team-game row using validated (team, season) keys

2. Feature engineering

- Build target variables:
 - Win (binary classification)
 - Point differential (regression)
- Create spending features: raw dollars by group, % allocation by group, “top-heavy” indicators, and year-over-year changes
- Construct derived ratios such as offense-to-defense spending and spending concentration indices

3. Machine learning with scikit-learn

- Regression: predict game-level point differential
- Classification: predict probability of winning a game
- Use time-aware train/test splits (e.g., training on earlier seasons, testing on later seasons) to prevent data leakage.
- Evaluate using regression metrics (MAE, RMSE) and classification metrics (accuracy, precision, recall, F1 score)

4. Final deliverable: interactive dashboard

- Exploratory views: spending vs outcomes, team comparisons, year slider, position-group breakdowns
- Model views: feature importance / coefficients, prediction vs actual, residual patterns
- Interactive prediction tool allowing users to adjust spending allocations and observe predicted outcomes

Required Deliverables

1. Machine Learning Models: regression model, classification model, evaluation metrics, brief interpretation

2. Data Pipeline: Reproducible code that loads CSV files, cleans, merges, and outputs a final modeling-ready dataset

3. Final Deliverable: Interactive dashboard or web app with real-time insights

4. GitHub Repository: Complete codebase with documentation and version control

Recommended Technical Stack

Python: pandas, numpy

Modeling: scikit-learn

Visualization: matplotlib, plotly

Dashboard: Panel

Version Control: Github

Why This Project?

This project stays strongly aligned with sports analytics while producing a portfolio-ready pipeline and app. Salary allocation is a real front office decision, and pairing it with game outcomes allows us to evaluate how financial strategy translates into win probability and performance margins. By quantifying how financial strategy translates to competitive outcomes, this project mirrors real-world sports analytics decision-making environments. The interactive dashboard makes the results easy to explore and present—perfect for a final deliverable and future coop conversations.